MACHINE LEARNING REPORT

ShopSmart Retail Ltd.

ML01 – Building Your First Machine Learning Models

This report documents data preparation, model development, evaluation and business interpretation for models that estimate totals and identify orders associated with product returns.

Models Developed:

1. Linear Regression - predicts **Order_Total**

2. Logistic Regression - predicts **Return_Flag**

Business Problem

ShopSmart Retail Ltd. wants to begin integrating Artificial Intelligence into its retail operations. Before developing advanced AI solutions, the company has hired AnalystLab Africa to build its first machine learning prototypes using historical retail data. As a Machine Learning Engineer, your task is to understand the fundamentals of machine learning, prepare datasets, build baseline models, and evaluate their performance.

Data Preparation

Nine business datasets were loaded: customers, employees, inventory, orders, order items, products, returns, stores and suppliers. A data dictionary was also available. Orders and order items were merged using order and product identifiers, with a one-to-one validation for transaction rows. Customer, product and store attributes were joined into a 50,00-row, 37-column master dataset. Employee, inventory and supplier data were retained separately because a reliable direct transaction-level join was not established in the workflow.

Line_Total was calculated as Quantity x Unit_Price – Discount. For this assignment, Order_Total was defined as Line_Total + Shipping_Cost. This is a project assumption because no specific formula was supplied. Return_Flag was set to 1 when an order ID appeared in the returns data and 0 otherwise. The returns table had 2,500 rows and 2,442 unique order IDs.

The data-quality checks recorded no missing cells or exact duplicate rows in the merged master dataset. Date fields were converted to datetime. The data was split into 80% training and 20% test sets, stratified by Return_Flag. Categorical features were one-hot encoded; numerical features were standerdised. The encoder and scaler were fitted on training data and applied to test data.

The selected features included Customer_Segment, Payment_Method, Sales_Channel, Category, Region, Quantity, Unit_Price, Discount and Shipping_Cost. The engineered regression model also used Quantity_Price = Quantity x Unit_Price. The required CSV outputs are data/master_dataset.csv, data/train.csv and data/test.csv.

Model Development

Linear Regression

A baseline Linear Regression model was trained to predict Order_Total using encoded categorical variables and four numerical variables.

An engineered version added Quantity_Price. The model was fitted on the transformed training matrix and evaluated on the held-out test matrix.

Logistic Regression

A Logistic Regression model was trained to predict Return_Flag. Since returned orders were a minority class, a standard model was compared with a class-weighted model using class_weight='balanced'. The class-weighted model assigned greater importance to the minority class.

The models and preprocessing objects were saved with Joblib. Both saved models were loaded and generated 10,000 predictions each. The loaded predictions matched the original models' predictions on the same prepared test matrices, verifying successful saving and reloading.

Model Evaluation

Regression

Model	MAE	RMSE	R^2
Baseline Linear Regression	29,538.14	40,079.22	0.8875
Engineered Linear Regression	Approximately 0	Approximately 0	1.0000

The baseline model explained a large proportion of test-target variation but its errors remained substantial. The engineered model achieved near-zero MAE and RMSE. $R^2 = 1.0000$ and a maximum absolute prediction error of approximately 2.62×10^{-10}.

Important: The engineered result reflects target-formula reconstruction because Quantity_Price and other transaction components are used to construct Order_Total. It should not be reported as proof of perfect future forecasting.

Classification

Metric	Baseline Logistic Regression	Class-weighted Logistic Regression
Accuracy	0.9512	0.5182
Returned-class precision	0.0000	0.0463 (4.6%)
Returned-class recall	0.0000	0.4529 (45.3%)
Returned-class F1	0.0000	0.0844 (8.4%)

The baseline accuracy of 95.12% is misleading: it predicted every order as not returned and detected no returned orders. The class-weighted model's confusion matrix was TN = 4,961, FP = 4,551, FN = 267, TP = 221. It detected some returns but many flagged orders were not returned. Its low precision means that operational use would create many false alerts.

Business Insights

1. Order Value

Quantity_Price dominated the engineered regression coefficients, followed by shipping cost and discount. This is consistent with the target formula, so it is primarily evidence of formula reconstruction rather than an independent finding about future customer spending.

2. Return-associated features

The largest absolute coefficients in the class-weighted logistic model included Region_East (+0.0849), Category_Home (+0.0789), Region_North (-0.0730), Category_Electronics (-0.0466), Quantity (+0.0406) and Shipping_Cost (-0.0370). Positive coefficients are associated with higher return scores and negative coefficients with lower scores, holding other features constant. They do not prove causation and should be treated cautiously given low precision.

3. Customer Segments

Silver Customers had the highest mean order value (about 151,835.92), Gold customers generated the most orders (16,961) and Bronze customers had the highest median order value (about 119,671). Mean values were close and the calculation used line totals excluding shipping.

4. Store Performance

ShopSmartLagos (STR015) had the highest total line value among the reported stores (311,890,052) and 2,064 orders. ShopSmartLagos (STR003) had the highest average line value among the displayed top stores (about 155,271.75). Store IDs should be used because multiple IDs can share a store name. Sales value not the same as profit.

5. Product variability

Within product standard deviation of line totals was used as a rough variability proxy. Examples included Home Item 95 (standard deviation about 92,369; 98 records), Groceries

Item 200 (about 94,577.59; 93 records). Lower variability does not necessarily imply better prediction accuracy; held-out errors should be measured by product.

Recommendations

- **Redesign the order-value target.** For genuine forecasting, predict future spending or order value using only information available before the outcome is known; avoid features that directly recreate the target formula.
- **Improve return-risk modelling.** Evaluate classification thresholds and alternative approaches using precision, recall, F1 and precision-recall trade-offs. Select thresholds based on the costs of missed returns and unnecessary reviews.
- **Use the classifier only as decision support.** Given its low precision, test any review process in a controlled pilot rather than automatically acting on every flagged order.
- **Collect richer data.** Useful additions may include purchase history, order frequency, customer tenure, return reasons, product defects, delivery delays, promotions, reviews, stock availability and store-level costs or profit.
- **Validate over time.** A time-based holdout and ongoing monitoring can show whether performance generalizes to future periods.
- **Keep preprocessing consistent.** For new data, preserve feature order and use the same encoder, scaler and engineered-feature definitions as in training.
- **Report business measures carefully.** Distinguish sales value from profit association from causation and formula reconstruction from predictive performance.

Limitations

- **Target construction:** Order_Total was derived from transaction fields used as predictors, limiting the meaning of the near-perfect regression result.
- **Class imbalance:** returned orders account for about 4.9% of records; accuracy alone conceals poor detection of the minority class.
- **Low precision:** the class-weighted classifier creates many false positives and has a low F1 score for returned orders.
- **Limited features:** the available modelling features omit potentially useful customer history, delivery experience, product quality, promotions and return reasons.
- **Model simplicity:** Linear and Logistic Regression may not capture nonlinear relationships or complex interactions.
- **Evaluation scope:** a stratified random split does not guarantee performance on future periods or under changing business conditions.

- **Interpretability:** model coefficients represent associations, not causal effects.